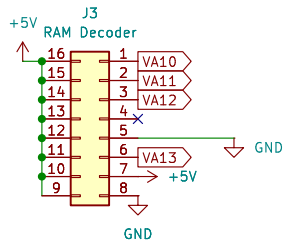
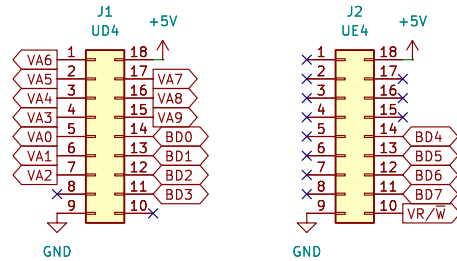


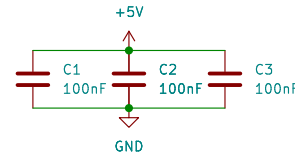
Signals to/from RAM decoder (74LS138)  
Original decoder activates internal RAM only for RAM0 and RAM4 to RAM5  
Now greatly simplified – we essentially don't need to care about the RAMx signals at all, just pull all the original 74LS138 outputs high.  
But we do need to use the incoming signals.



Original 2114 RAM chips are removed from the VIC.  
Two are replaced with IC sockets so daughterboard can be installed.

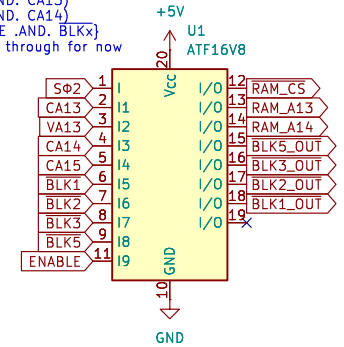


#### Power & Decoupling



Remove lots of different logic gates with a single PLD  
We need to handle whether CPU or VIC needs to access the RAM  
When SΦ2 is low, CPU is accessing RAM and CA13/CA14 are valid  
When SΦ2 is high, VIC is accessing RAM and CA13/CA14 must be low as VIC only reads BLK0  
If expansion is disabled then route BLK1/BLK2/BLK3 back to motherboard

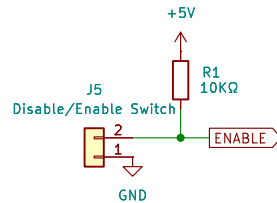
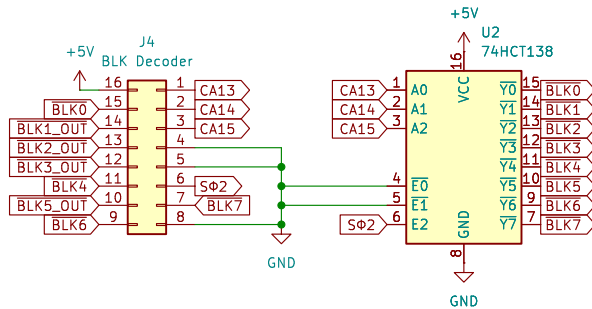
RAM\_CS = (SΦ2 .AND. CA15) .OR. (SΦ2 .AND. VA13)  
RAM\_A13 = (SΦ2 .AND. CA13)  
RAM\_A14 = (SΦ2 .AND. CA14)  
BLKx\_OUT = (ENABLE .AND. BLKx)  
BLK5 is just passed through for now



#### Signals to/from BLK decoder (74LS138)

When our expansion is active we want to prevent BLK1 to BLK3 selects returning to the VIC  
But we also want to be able to disable the expansion for software compatibility  
This can be done via a switch:

\* When the expansion is disabled the BLK1/BLK2/BLK3 (and possibly BLK5 in future) are passed back to the VIC, disconnecting our RAM expansion from the data bus and passing through to the cartridge port  
\* When the expansion is enabled, these are blocked which prevents the VIC from disconnecting our RAM from the data bus

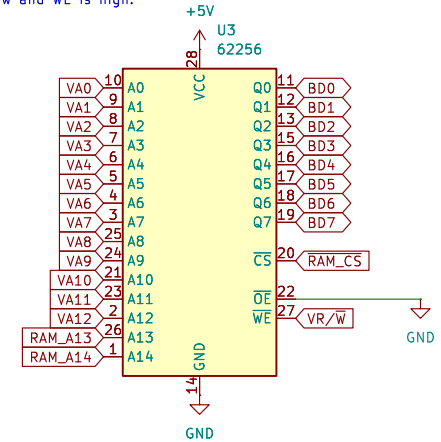


#### 32KB RAM Expansion

Read: CS low & VR/W high → OE low

Write: CS low & VR/W low → OE high

Dave's comment: The internal logic will take over if OE is tied low. It will only output if CS is low and WE is high.



Future:  
To support BLK5 expansion we will need to add an additional 6264

With lots of help from Dave Curran @ Tynemouth Software

Refer to the "DIP-16 Breakout Board" for the RAM/BLK decoder connection

Brett Hallen

Sheet: /

File: VIC-20\_Internal\_32KB\_RAM.kicad\_sch

Title: Commodore VIC-20 32KB RAM Expansion

Size: A4

Date: 11/JUN/2026

Rev: B

KiCad E.D.A. 9.0.9

Id: 1/1